Code No: 135CH **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
FUNDAMENTALS OF MECHANICAL ENGINEERING
(Common to CE, EEE, CSE, MIE)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the significance of heat transfer, work transfer and internal energy.
b) State and explain the first law of thermodynamics. [8+7]

2. Explain in detail the reasons for ozone depletion and suggest some remedies to mitigate the effect of global warming. [15]

3. Explain the following terms:
a) Specific volume
b) Dryness fraction of steam
c) Steam calorimeters. [5+5+5]

4. Explain adiabatic process and Poly-tropic processes and discuss their applications. [15]

5. Explain how do you describe the thermal efficiency of Rankine cycle and derive its equation from P-V and T-S diagrams. [15]

6. Explain the working principle of four- stroke Diesel engines with a suitable diagram. [15]

7.a) Explain the working of vapor absorption refrigeration system with the help of a neat sketch.
b) Explain the working of centrifugal pumps with their characteristics. [8+7]

8.a) Explain the construction and applications of universal couplings.
b) How does the shoe and disc brake work? Explain. [8+7]

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